

Descriptive Statistics for Market Data

Market-data summaries, distribution shape, timing, weighting, and production metrics.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0017
CATEGORY	Statistics
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

Table of contents

BEGINNER

ORIENTATION

Title	1
How to use this manual	4
Notation and measurement contract	5
Market-data summary workflow	6

FOUNDATIONS

Observation, variable, unit, and grain	7
Price levels versus returns	8
Simple, log, and cumulative returns	9
Frequency, horizon, and annualization	10
Population, sample, and eligibility	11
Mean, median, and mode	12
Weighted and grouped averages	13
Range, variance, and standard deviation	14
Standard scores and normalization	15

DISTRIBUTION SHAPE

Quantiles and percentiles	16
Five-number summary and IQR	17
Skewness and directional asymmetry	18
Kurtosis and tail concentration	19
Median absolute deviation and robust scale	20
Trimming, winsorization, and influence	21
Empirical distribution and ECDF	22
Histograms and binning choices	23
Kernel density and smoothing	24

Table of contents

BEGINNER

TIME-SERIES SUMMARIES

Autocorrelation as a descriptive pattern	25
Rolling windows and local statistics	26
Expanding windows and cumulative evidence	27
Exponentially weighted statistics	28
Realized volatility and range estimators	29
Drawdown depth, duration, and recovery	30
Downside, upside, and semideviation	31
Seasonality and intraday profiles	32
Regime-conditioned summaries	33

CROSS-SECTIONAL AND SYSTEMS

Cross-sectional ranks and percent ranks	34
Breadth and cross-sectional dispersion	35
Correlation and covariance summaries	36
Missingness and coverage summaries	37
Outliers and influence diagnostics	38
Grouped summaries and Simpson's paradox	39
Descriptive metric contracts	40
Descriptive dashboards and monitoring	41

VISUAL LABS AND REVIEW

From raw prices to valid summaries	42
Reading a market-return distribution	43
Rolling, expanding, and exponential memory	44
Conditional performance and mixture risk	45
Choosing a descriptive view	46
Formula sheet and interpretation map	47
Worked calculation: return sample	48
Failure modes and diagnostic actions	49
Final active recall and system index	50

How to use this manual

BEGINNER

1. DEFINE THE ROW

Write entity, timestamp, horizon, unit, and eligibility before touching a summary function. This prevents a correct calculation from answering the wrong market question.

2. INSPECT CENTER AND SPREAD TOGETHER

Pair an economic center with ordinary and robust dispersion. Mean without tails and volatility without location each hide a different part of the distribution.

3. PRESERVE TIME AND CROSS-SECTION

Decide whether the reference set is one asset through time or many assets at one time. Never blend these interpretations under a generic normalization label.

4. LOOK BEYOND THE AGGREGATE

Use quantiles, regimes, breadth, group weights, and contribution diagnostics to find concentration, mixture effects, and valid extremes.

5. MAKE TRANSFORMATIONS POINT-IN-TIME

Lag rolling features, version annualization, retain raw observations, and fit clipping or scaling boundaries only from information available at the decision timestamp.

6. OPERATIONALIZE THE DEFINITION

Turn important summaries into metric contracts with grain, formula, units, weights, windows, tests, lineage, owner, and quality-state behavior.

READING PATTERN

Each concept page moves from meaning to market use, equation intuition, worked example, implementation, failure mode, system placement, and active recall.

Notation and measurement contract

BEGINNER

SYMBOL	OBJECT	MEANING
P_t	price	Adjusted or raw price at time t , explicitly identified
r_t	simple return	$P_t/P_{t-1}-1$ over a declared horizon
ell_t	log return	$\ln(P_t/P_{t-1})$, additive through time
$\bar{x}$	sample mean	Equal-row average over eligible observations
x_{tilde}	sample median	Middle ordered value under a pinned convention
s	sample scale	Standard deviation using stated degrees of freedom
Q_q	quantile	Value at cumulative fraction q
IQR	robust spread	$Q_{0.75}$ minus $Q_{0.25}$
MAD	robust scale	Median absolute distance from the median
w_i	weight	Declared economic influence of observation i

ALWAYS ATTACH

Entity and timestamp grain, source field, adjustment basis, currency or percent unit, sampling frequency, economic horizon, eligibility, weights, grouping, window, lag, null policy, and count.

CORE DISTINCTION

A statistic is not only a formula. The same function applied to trade rows, asset-days, value-weighted assets, or regime-filtered dates describes different economic populations.

Market-data summary workflow

BEGINNER

1

Frame the question

Decision, population, entity, clock, horizon, and units

2

Validate observations

Corporate actions, timestamps, duplicates, staleness, and coverage

3

Choose weighting

Equal asset, traded dollar, exposure, event, or fixed reference mix

4

Summarize location

Mean, median, quantiles, and economically relevant thresholds

5

Describe shape

Dispersion, asymmetry, tails, empirical distribution, and influence

6

Respect structure

Time dependence, seasonality, regimes, groups, and cross-section

7

Stress interpretation

Robust variants, contribution analysis, and alternative windows

8

Publish the contract

Definition, lineage, tests, owner, freshness, and monitoring action

HARD GATE

No descriptive number reaches a model, limit, or dashboard until its denominator, timing, transformation, and data-quality behavior can be reproduced from a sealed snapshot.

Observation, variable, unit, and grain

BEGINNER

ONE-SENTENCE MEANING

An observation is one measured record, a variable is a field recorded on it, a unit states the measurement scale, and grain defines what one row represents. Descriptive statistics are valid only after all four are explicit.

MEMORY JOGGER

Before computing an average, finish this sentence: one row is one asset, at one timestamp, for one horizon, measured in one unit. Ambiguity at the row level becomes ambiguity in every summary.

WHY IT MATTERS IN QUANT RESEARCH

Market tables mix trades, quotes, bars, positions, and events. A mean across incompatible grains can overweight active names, duplicate information, or combine percentages with currency amounts.

FORMULA INTUITION

A summary has the form $S(\{x_i\}, \text{eligibility, weights, grouping})$. The numerical values alone are insufficient because membership, weighting, and aggregation determine what S means.

SIMPLE MARKET EXAMPLE

A table with one row per trade gives busy stocks more influence than a table with one row per stock-day. Both can report average spread, but they answer different operational questions.

PYTHON / SYSTEM IMPLEMENTATION

Attach schema metadata for entity key, event time, effective time, timezone, currency, scale, horizon, and primary key. Assert uniqueness at the declared grain before aggregation.

CONFUSIONS TO AVOID

Do not infer grain from column names or assume duplicate timestamps are errors; multiple venues can legitimately print simultaneously. Resolve economic identity before deduplication.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data contract establishes grain at ingestion, feature jobs preserve it through transformations, and every research metric records its input grain so downstream comparisons remain auditable.

RELATED CONCEPTS AND QUICK RECALL

Related: data schema, panel data, aggregation, measurement scale. Recall task: State the grain and unit for a daily close-to-close return panel, then explain how trade-level rows would change its weighting.

Price levels versus returns

BEGINNER

ONE-SENTENCE MEANING

A price level is the current scale of an asset, while a return expresses change relative to a prior value. Prices are usually nonstationary and incomparable across assets; returns are closer to a common economic unit.

MEMORY JOGGER

A \$200 stock is not twice as expensive as a \$100 stock in an investment sense. Compare proportional change, not the arbitrary scale created by shares outstanding and split history.

WHY IT MATTERS IN QUANT RESEARCH

Most performance, volatility, dependence, and signal summaries should operate on adjusted returns rather than raw prices. Price-level summaries remain useful for execution constraints, tick size, and notional calculations.

FORMULA INTUITION

Simple return $r_t = P_t / P_{t-1} - 1$. Dollar change is $\Delta P_t = P_t - P_{t-1}$. The first is scale free; the second preserves currency exposure and must be paired with position size.

SIMPLE MARKET EXAMPLE

Asset A rises from 20 to 21 and Asset B from 200 to 202. B has the larger dollar move, but A earns 5% versus B's 1%, so a \$10,000 investment performs better in A.

PYTHON / SYSTEM IMPLEMENTATION

Use split- and dividend-adjusted prices for economic returns, but retain raw prices for execution simulation. Validate return reconstruction around corporate actions against vendor total-return fields.

CONFUSIONS TO AVOID

Averaging price levels across assets has little economic meaning, and calculating returns from unadjusted closes creates artificial jumps. Adjustment policy must match the research objective.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The canonical market-data layer stores raw, adjusted, and total-return series separately. Feature definitions declare which one they consume, preventing execution fields from leaking into performance accounting.

RELATED CONCEPTS AND QUICK RECALL

Related: simple return, total return, corporate actions, stationarity. Recall task: Explain why a stock split can corrupt price-change statistics while leaving correctly adjusted returns unchanged.

Simple, log, and cumulative returns

BEGINNER

ONE-SENTENCE MEANING

Simple returns measure percentage gain over a period, log returns measure the logarithm of the price ratio, and cumulative returns compound a sequence into total wealth change. Each representation supports different operations.

MEMORY JOGGER

Simple returns compound by multiplication; log returns aggregate by addition. Small one-period values are similar, but the difference matters for large moves and portfolio aggregation.

WHY IT MATTERS IN QUANT RESEARCH

Return choice affects descriptive means, volatility scaling, forecasting targets, and reconciliation with realized portfolio P&L. A metric registry should never label all three merely as return.

FORMULA INTUITION

$$r_t = P_t / P_{t-1} - 1; \text{ ell}_t = \ln(P_t / P_{t-1}) = \ln(1 + r_t); R_{1:T} = \text{product}(1 + r_t) - 1 = \exp(\text{sum ell}_t) - 1.$$

SIMPLE MARKET EXAMPLE

Returns of +20% then -20% have arithmetic mean 0%, but wealth becomes 0.96 and cumulative return is -4%. Log returns sum to $\ln(0.96)$, preserving the actual path endpoint.

PYTHON / SYSTEM IMPLEMENTATION

Compute with explicit sort order and entity partitions. Use `log1p(r)` and `expm1(sum_log)` for numerical stability, and fail on prices that are nonpositive or stale across invalid gaps.

CONFUSIONS TO AVOID

Log returns do not aggregate across assets using portfolio weights the way simple returns do. Arithmetic average return is not compound growth, and cumulative return is not a sum of percentages.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The return service publishes named fields for simple, log, excess, and total returns plus the source price policy. Backtests reconcile compounded returns to the equity curve on every run.

RELATED CONCEPTS AND QUICK RECALL

Related: geometric mean, compounding, wealth index, excess return. Recall task: Calculate the cumulative return for +10%, -5%, and +3%, and contrast it with their arithmetic sum.

Frequency, horizon, and annualization

BEGINNER

ONE-SENTENCE MEANING

Frequency states how often observations are sampled, while horizon states the interval each value covers. Annualization rescales a statistic under assumptions that must be appropriate for that statistic and process.

MEMORY JOGGER

Daily data and one-day returns are not interchangeable descriptions: daily rows can contain 20-day forward returns. Always name both sampling clock and economic horizon.

WHY IT MATTERS IN QUANT RESEARCH

Comparing strategies requires common units, but naive square-root scaling can misstate risk when returns are autocorrelated, volatility changes, or calendars differ. Horizon alignment also prevents label overlap from masquerading as evidence.

FORMULA INTUITION

For IID returns, annualized mean is k times the period mean and annualized volatility is $\sqrt{k}$ times period volatility. Compounded annual return is $\text{product}(1+r_t)$ over the year minus one.

SIMPLE MARKET EXAMPLE

A daily volatility of 1% implies about 15.9% annualized volatility under 252 independent days. Positive autocorrelation raises long-horizon variance above this shortcut.

PYTHON / SYSTEM IMPLEMENTATION

Store calendar ID, observations-per-year convention, session boundary, and return horizon. Compute exact elapsed-time and realized compounding where possible, then label approximations explicitly.

CONFUSIONS TO AVOID

Do not use 252 for every market, multiply Sharpe by $\sqrt{252}$ when returns overlap, or annualize a five-day statistic as if it were one-day. Scaling is a model, not formatting.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric layer centralizes annualization conventions and exposes unannualized values for reconciliation. Validation compares direct long-horizon statistics with scaled short-horizon estimates.

RELATED CONCEPTS AND QUICK RECALL

Related: sampling frequency, overlapping returns, calendar, time aggregation. Recall task: List the assumptions behind converting daily volatility to annual volatility and name one diagnostic for each assumption.

Population, sample, and eligibility

BEGINNER

ONE-SENTENCE MEANING

The population is the full set of market observations relevant to a decision, the sample is the observed subset summarized, and eligibility rules determine which records can enter. Descriptions inherit the scope of that selection.

MEMORY JOGGER

A statistic describes its sample first, not the market in general. Generalization requires a reason that the included assets, dates, and regimes represent the intended deployment population.

WHY IT MATTERS IN QUANT RESEARCH

Universe screens, listing histories, vendor coverage, and quality filters can reshape average returns, spreads, and tails. Descriptive work should reveal selection rather than bury it in a query.

FORMULA INTUITION

For eligible indicator e_i , a mean is $\text{sum}(e_i \times x_i) / \text{sum}(e_i)$. Changing e_i changes the estimand even when the observed values and code for averaging stay the same.

SIMPLE MARKET EXAMPLE

Summarizing today's index constituents over ten years excludes firms removed after poor performance. The resulting return and drawdown profile describes survivors, not the historical index opportunity set.

PYTHON / SYSTEM IMPLEMENTATION

Build point-in-time eligibility snapshots and publish included and excluded counts by date and reason. Hash the rule set so a dashboard value can be reproduced from the same membership contract.

CONFUSIONS TO AVOID

Dropping nulls silently is an eligibility decision, not harmless cleanup. A large sample can still be systematically biased if the frame omits delisted, illiquid, or interrupted observations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Universe construction precedes feature computation. The evidence catalog binds every summary to universe version, cutoff, quality state, and exclusion ledger before it reaches research notebooks.

RELATED CONCEPTS AND QUICK RECALL

Related: sampling frame, survivorship bias, missingness, external validity. Recall task: Define a defensible population for studying US equity closing-auction spreads and identify three eligibility fields.

Mean, median, and mode

BEGINNER

ONE-SENTENCE MEANING

The mean balances all values, the median splits ordered observations in half, and the mode identifies the most frequent value or category. They summarize different notions of center and respond differently to skew and discreteness.

MEMORY JOGGER

Mean follows dollars and tails; median follows the typical ranked observation; mode follows concentration. Ask which interpretation matches the decision before choosing one.

WHY IT MATTERS IN QUANT RESEARCH

Average trade return supports expected P&L, median return describes a typical trade, and modal spread can expose price-grid clustering. Reporting multiple centers reveals asymmetry that one number conceals.

FORMULA INTUITION

Mean= $\bar{x} = \sum x_i / n$. Median is the middle order statistic, averaging the two middle values for even n . A numeric mode depends on exact ties and is often unstable for continuous values.

SIMPLE MARKET EXAMPLE

Returns of -8%, 0%, 1%, 1%, and 2% have mean -0.8%, median 1%, and mode 1%. The strategy's typical trade is positive while one loss makes expected return negative.

PYTHON / SYSTEM IMPLEMENTATION

Compute centers on the same eligible sample and display count plus tail quantiles beside them. For floating-point modes, use economically meaningful bins such as tick or basis-point increments.

CONFUSIONS TO AVOID

The median is not a robust substitute for every objective: a portfolio earns the sum of P&L, not the median trade. A mode from arbitrary histogram bins is partly a plotting choice.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research scorecards pair economic mean with median, hit rate, and tail contribution. Alerting highlights sign disagreement between mean and median because it often signals concentrated downside or data error.

RELATED CONCEPTS AND QUICK RECALL

Related: location, order statistic, expected value, skewness. Recall task: Construct five returns with a positive median and negative mean, then explain which metric governs capital growth.

Weighted and grouped averages

BEGINNER

ONE-SENTENCE MEANING

A weighted average gives observations different influence according to a declared economic rule, while a grouped average summarizes within strata before combining. The weights define whose experience the statistic represents.

MEMORY JOGGER

Every average has an implicit weighting scheme. Equal-row weighting is a choice, not the absence of a choice.

WHY IT MATTERS IN QUANT RESEARCH

Market summaries may be equal-weighted by asset, value-weighted by capitalization, volume-weighted by trading activity, or capital-weighted by exposure. These answer materially different questions.

FORMULA INTUITION

Weighted mean $\bar{x}_w = \frac{\sum(w_i x_i)}{\sum(w_i)}$. To combine group means, use group mass in the target population rather than averaging group means equally unless equal groups are the goal.

SIMPLE MARKET EXAMPLE

A 10 bp spread in a stock trading \$1 million and a 2 bp spread in one trading \$100 million average to 6 bp equally but about 2.08 bp when dollar-volume weighted.

PYTHON / SYSTEM IMPLEMENTATION

Store weight field, units, timestamp, normalization, clipping policy, and missing-weight behavior. Reconcile aggregate weighted sums to raw numerator and denominator totals.

CONFUSIONS TO AVOID

Using future market capitalization creates look-ahead bias, and volume weighting can overrepresent stressed trading. Negative portfolio weights require different interpretation from survey-style positive weights.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric contract names the economic perspective, such as asset opportunity, traded dollar, or deployed capital. Dashboards allow comparison only when weight semantics match.

RELATED CONCEPTS AND QUICK RECALL

Related: VWAP, equal weighting, market capitalization, Simpson's paradox. Recall task: Compare equal-weighted and dollar-volume-weighted spread estimates and state the decision each supports.

Range, variance, and standard deviation

BEGINNER

ONE-SENTENCE MEANING

Range records the distance between extremes, variance averages squared deviations from the mean, and standard deviation expresses that dispersion in the original unit. They capture spread but not the full distribution shape.

MEMORY JOGGER

Range listens only to two points; variance gives extreme deviations quadratic influence; standard deviation translates variance back into readable units.

WHY IT MATTERS IN QUANT RESEARCH

Volatility, tracking error, feature scaling, and risk budgets rely on standard deviation. Comparing it with robust scale and quantiles reveals whether a few events dominate the estimate.

FORMULA INTUITION

Sample variance $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$; standard deviation $s = \sqrt{s^2}$; range $= \max(x) - \min(x)$. The $n-1$ correction does not make heavy tails or dependence harmless.

SIMPLE MARKET EXAMPLE

For returns -1%, 0%, 1%, 2%, and 8%, the range is 9 percentage points. The 8% observation contributes far more to variance than its one-fifth row share.

PYTHON / SYSTEM IMPLEMENTATION

Calculate in stable two-pass or vetted library routines, partitioned by asset and horizon. Record degrees of freedom and compare with MAD, IQR, and leave-one-out sensitivity.

CONFUSIONS TO AVOID

Standard deviation is not downside risk and does not say how often extremes occur. Mixing percentages and decimals can create a 100-fold scale error that still looks statistically valid.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk-feature service publishes raw and robust dispersion together, with units and window length. Data-quality monitors flag abrupt scale ratios before models consume them.

RELATED CONCEPTS AND QUICK RECALL

Related: volatility, semivariance, MAD, tail risk. Recall task: Explain why two return samples can share the same standard deviation yet have very different trading risk.

Standard scores and normalization

BEGINNER

ONE-SENTENCE MEANING

A standard score expresses an observation as deviations from a reference center measured in reference scale units. Normalization makes features comparable, but its meaning depends on the chosen window, group, and robust estimators.

MEMORY JOGGER

A z-score is a location statement relative to a reference set, not a probability statement unless a distribution model justifies that translation.

WHY IT MATTERS IN QUANT RESEARCH

Standardized signals support cross-asset ranking, anomaly detection, and model inputs. Point-in-time fitting is essential because using future mean and scale leaks information into historical features.

FORMULA INTUITION

Classical $z=(x-\mu)/\sigma$; sample $z=(x-\bar{x})/s$. A robust alternative is $z_r=(x-\text{median})/(1.4826 \text{ MAD})$ when a roughly normal reference makes that consistency factor useful.

SIMPLE MARKET EXAMPLE

A 3% daily move is ordinary for a volatile small-cap but exceptional for a stable bond future. Group-specific rolling standardization reflects the relevant baseline.

PYTHON / SYSTEM IMPLEMENTATION

Fit location and scale only on data available before the timestamp, persist parameters, enforce minimum observations, and cap or flag near-zero denominators. Separate cross-sectional from time-series normalization.

CONFUSIONS TO AVOID

A z-score of 3 does not automatically imply a 0.3% tail probability in heavy-tailed returns. Global scaling can also erase regime changes or encode asset identity.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature pipeline versions scaler scope, window, lag, robust method, and fallback. Live monitoring compares input z-score distributions with training expectations without recomputing history.

RELATED CONCEPTS AND QUICK RECALL

Related: feature scaling, robust z-score, leakage, cross-sectional rank. Recall task: Contrast a time-series z-score with a same-day cross-sectional z-score for the same asset return.

Quantiles and percentiles

BEGINNER

ONE-SENTENCE MEANING

A quantile is a value below which a stated fraction of observations falls; a percentile expresses that fraction on a 0-to-100 scale. Quantiles describe distribution position without assuming symmetry or normality.

MEMORY JOGGER

Sort first, then ask where an observation sits. Quantiles are about rank and empirical probability mass, not distance from the mean.

WHY IT MATTERS IN QUANT RESEARCH

Return tails, spread limits, turnover budgets, and latency service levels are naturally described by quantiles. They remain interpretable when averages are distorted by skew or occasional extremes.

FORMULA INTUITION

The empirical q -quantile is an order-statistic interpolation $x_{(q)}$. Software packages use several interpolation conventions, so small-sample values can differ even on identical data.

SIMPLE MARKET EXAMPLE

If daily slippage has a 95th percentile of 12 bp, about 5% of observed eligible days were worse than 12 bp. It does not promise the next 5% will follow the same pattern.

PYTHON / SYSTEM IMPLEMENTATION

Pin the quantile algorithm, sample scope, weights, and minimum count. Use approximate streaming quantiles only with documented error bounds and regression tests against exact calculations.

CONFUSIONS TO AVOID

The 99th percentile is unstable with a few hundred rows, and quantiles do not extrapolate beyond observed support. Calling the 5th return percentile a 95% VaR also requires a loss-sign convention.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metrics service publishes a consistent percentile grid with count and coverage metadata. Risk limits use conservative interpolation and retain the underlying tail observations for audit.

RELATED CONCEPTS AND QUICK RECALL

Related: order statistics, empirical CDF, VaR, rank. Recall task: How many independent observations support an empirical 99th percentile, and what makes that estimate fragile?

Five-number summary and IQR

BEGINNER

ONE-SENTENCE MEANING

The five-number summary contains minimum, first quartile, median, third quartile, and maximum; the interquartile range is $Q3 - Q1$. Together they provide a compact map of center, central spread, and extremes.

MEMORY JOGGER

The middle box describes half the data; the whisker endpoints describe observed reach. Large asymmetry on either side is visible before any higher moment is calculated.

WHY IT MATTERS IN QUANT RESEARCH

Five-number summaries quickly compare returns, spreads, signal values, and holding times across assets or regimes. IQR is a robust scale measure for screening without giving extremes quadratic weight.

FORMULA INTUITION

$IQR = Q_{0.75} - Q_{0.25}$. Tukey fences mark values below $Q1 - 1.5 \cdot IQR$ or above $Q3 + 1.5 \cdot IQR$ as inspection candidates, not automatic errors.

SIMPLE MARKET EXAMPLE

Daily returns with $Q1 = -0.6\%$, $median = 0.05\%$, and $Q3 = 0.7\%$ have $IQR = 1.3\%$. A -4% day lies far below the central range but may still be a genuine market event.

PYTHON / SYSTEM IMPLEMENTATION

Compute with a pinned quantile definition and display counts. Store fence flags separately from cleaning decisions so researchers can inspect whether flagged observations cluster around known events.

CONFUSIONS TO AVOID

Boxplot outliers are not statistical impossibilities or bad ticks. Min and max grow more extreme with sample size, making cross-group comparisons misleading when counts differ greatly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploration dashboards pair five-number summaries with observation counts, date coverage, and raw-event drilldowns. Quality workflows route flagged points to validation rather than deleting them.

RELATED CONCEPTS AND QUICK RECALL

Related: boxplot, robust scale, quantile, outlier. Recall task: Explain why a valid crash return can cross a Tukey fence and still belong in a risk summary.

Skewness and directional asymmetry

BEGINNER

ONE-SENTENCE MEANING

Skewness measures imbalance between the left and right sides of a distribution, emphasizing the direction and leverage of large deviations. Negative return skew often signals occasional losses that dominate many small gains.

MEMORY JOGGER

Ask which tail stretches farther and carries more magnitude. Skewness converts that visual asymmetry into a standardized third-moment summary.

WHY IT MATTERS IN QUANT RESEARCH

Carry, short-volatility, and mean-reversion strategies can look stable by mean and standard deviation while hiding negative skew. Skew also affects confidence intervals and the interpretation of a typical outcome.

FORMULA INTUITION

Population skewness $\gamma_1 = E[(X - \mu)^3] / \sigma^3$. Finite-sample estimators apply bias corrections, but all remain highly sensitive to extreme observations.

SIMPLE MARKET EXAMPLE

A strategy earning +0.2% on most days and losing -5% occasionally can have a positive median and Sharpe yet strongly negative skew. The loss tail is the economic signature.

PYTHON / SYSTEM IMPLEMENTATION

Report the estimator, count, window, and bootstrap uncertainty. Compare moment skewness with $Q90 + Q10 - 2Q50$ scaled by IQR and inspect tail-contribution plots.

CONFUSIONS TO AVOID

Zero skewness does not imply a normal distribution, and sign can flip across regimes. A single bad tick can dominate cubic deviations, so lineage inspection precedes interpretation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy scorecard places skew beside drawdown, expected shortfall, and regime counts. Promotion rules escalate negative skew even when average return remains attractive.

RELATED CONCEPTS AND QUICK RECALL

Related: third moment, downside tail, asymmetry, tail contribution. Recall task: Describe a payoff with positive win rate, positive median, and negative skew, then name the risk control it needs.

Kurtosis and tail concentration

BEGINNER

ONE-SENTENCE MEANING

Kurtosis is the standardized fourth central moment and reflects how strongly extreme deviations contribute relative to variance. Excess kurtosis subtracts three so a normal distribution has value zero.

MEMORY JOGGER

Kurtosis is better remembered as extreme-event sensitivity than as peak height. The fourth power makes rare large moves dominate the statistic.

WHY IT MATTERS IN QUANT RESEARCH

Financial returns frequently exhibit fat tails, so normal-based risk estimates can understate shock frequency. Kurtosis also warns that volatility estimates and sample means may converge slowly.

FORMULA INTUITION

Kurtosis $\gamma_2 = E[(X-\mu)^4]/\sigma^4$; excess = $\gamma_2 - 3$. Sample formulas differ in finite-sample bias correction and should be versioned.

SIMPLE MARKET EXAMPLE

Two series can share 1% volatility, but one with occasional 8% moves will have much higher kurtosis and larger margin or stop-out risk than a smooth Gaussian-like series.

PYTHON / SYSTEM IMPLEMENTATION

Compute alongside tail quantiles, exceedance counts, and maximum contribution to the fourth moment. Use log-scale tail plots and bootstrap intervals because point estimates are unstable.

CONFUSIONS TO AVOID

High kurtosis does not identify which tail is dangerous and should not be called peakedness alone. Removing extremes to stabilize it may erase the very risk the metric detects.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk data product stores left- and right-tail exceedances with kurtosis, allowing reviewers to separate crash exposure from upside jumps and data defects.

RELATED CONCEPTS AND QUICK RECALL

Related: fourth moment, fat tails, expected shortfall, normality. Recall task: Why can a high-kurtosis return series have an ordinary-looking histogram center but dangerous risk behavior?

Median absolute deviation and robust scale

BEGINNER

ONE-SENTENCE MEANING

Median absolute deviation, or MAD, is the median distance from the sample median. It estimates typical dispersion while resisting a minority of extreme values.

MEMORY JOGGER

Center robustly, take absolute distances, then choose the middle distance. Unlike variance, an extreme point cannot grow its influence by being squared.

WHY IT MATTERS IN QUANT RESEARCH

MAD supports anomaly detection, robust feature scaling, and stable regime comparisons when market returns contain jumps. Comparing MAD with standard deviation is itself a tail diagnostic.

FORMULA INTUITION

$MAD = \text{median}(|x_i - \text{median}(x)|)$. For normal data, 1.4826 MAD estimates sigma, but that scaling encodes a distribution reference and is not universally correct.

SIMPLE MARKET EXAMPLE

If ordinary volatility is 2.0% and scaled MAD is 0.9%, a small number of large moves contribute heavily to variance. Both numbers are useful rather than one replacing the other.

PYTHON / SYSTEM IMPLEMENTATION

Publish raw MAD and any consistency-scaled version separately. Define behavior for short windows, ties, and zero MAD; fall back to IQR or a minimum scale only through an explicit rule.

CONFUSIONS TO AVOID

Robust does not mean immune: contamination near or above half the sample can shift the median. MAD can also be zero for discrete or stale data even when occasional moves are large.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Feature services use robust scale as an optional transform and monitor the standard-deviation-to-MAD ratio. Abrupt ratio changes trigger tail and data-quality investigation.

RELATED CONCEPTS AND QUICK RECALL

Related: robust statistics, IQR, z-score, breakdown point. Recall task: Interpret a rising standard-deviation-to-MAD ratio when median volatility stays stable.

Trimming, winsorization, and influence

BEGINNER

ONE-SENTENCE MEANING

Trimming removes a declared fraction of extreme observations, while winsorization replaces them with boundary values. Both limit influence but change the data representation and often the estimand.

MEMORY JOGGER

Trim deletes tail mass; winsorize caps tail magnitude. Neither is a substitute for determining whether an extreme value is valid.

WHY IT MATTERS IN QUANT RESEARCH

Robust summaries can keep noisy cross-sectional features from being driven by one asset, yet trading risk often lives in exactly those extremes. Raw and adjusted views should coexist.

FORMULA INTUITION

An alpha-trimmed mean averages observations between empirical alpha and 1-alpha quantiles. A winsorized mean averages values after clipping to those quantile boundaries.

SIMPLE MARKET EXAMPLE

Clipping the most extreme 1% of daily returns may stabilize a factor exposure estimate while severely understating a short-option strategy's true loss distribution.

PYTHON / SYSTEM IMPLEMENTATION

Fit boundaries point-in-time and within the intended group, log how many values change, and preserve original fields. Test sensitivity across thresholds rather than selecting the most favorable result.

CONFUSIONS TO AVOID

Symmetric clipping can distort naturally asymmetric data, and cross-sectional winsorization with few names is unstable. Never use cleaned returns for portfolio P&L reconciliation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature layer owns adjusted variants with explicit suffixes and lineage. The risk layer consumes unaltered outcomes and may use robust estimators only as complementary diagnostics.

RELATED CONCEPTS AND QUICK RECALL

Related: robust mean, outliers, clipping, influence function. Recall task: When is winsorization defensible for a predictor but unacceptable for realized strategy returns?

Empirical distribution and ECDF

BEGINNER

ONE-SENTENCE MEANING

The empirical cumulative distribution function, or ECDF, reports the fraction of observations less than or equal to each value. It preserves the full ranked sample without choosing histogram bins.

MEMORY JOGGER

For any threshold, the ECDF answers how much observed mass lies at or below it. Its complement answers exceedance frequency.

WHY IT MATTERS IN QUANT RESEARCH

ECDFs compare slippage, drawdown, latency, and return distributions across models or regimes. Vertical gaps reveal where one process dominates another rather than compressing differences into one mean.

FORMULA INTUITION

$F_n(x) = (1/n) \sum I(x_i \leq x)$. The empirical survival function is $1 - F_n(x)$, with careful convention at ties.

SIMPLE MARKET EXAMPLE

If $F_n(-2\%) = 0.04$, then 4% of observed returns were at or below -2%. A second strategy with 0.09 at the same threshold has more frequent losses beyond that level.

PYTHON / SYSTEM IMPLEMENTATION

Use exact sorted values for moderate data or mergeable sketches for scale. Preserve weights, ties, and eligibility, and show confidence bands or counts when comparing thin tails.

CONFUSIONS TO AVOID

ECDF steps are empirical, not a smooth population law. Curves can cross, so one strategy may be better around ordinary losses but worse in the extreme tail.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Distribution-comparison services store percentile grids and threshold exceedances from a common snapshot. Review interfaces allow drilldown from a curve segment to the underlying dates.

RELATED CONCEPTS AND QUICK RECALL

Related: CDF, survival function, stochastic dominance, quantile. Recall task: What does a vertical ECDF gap at -1% mean economically when comparing two strategies?

Histograms and binning choices

BEGINNER

ONE-SENTENCE MEANING

A histogram partitions numeric values into intervals and displays counts or density within each bin. It reveals shape quickly, but apparent modes, gaps, and tails can change with bin width and origin.

MEMORY JOGGER

A histogram is a lossy compression of sorted observations. Change the buckets and the visual story can change even though the data do not.

WHY IT MATTERS IN QUANT RESEARCH

Histograms expose discretization, stale prices, return spikes, multimodality, and regime mixtures. They are most useful when paired with summary statistics and a sensitivity view.

FORMULA INTUITION

For equal-width bins, density $\text{height} = \text{count} / (n \times \text{bin width})$, so total bar area equals one. Rules such as Freedman-Diaconis use $\text{width} = 2 \times \text{IQR} \times n^{-1/3}$ as a starting point.

SIMPLE MARKET EXAMPLE

Daily returns may look bell-shaped at 1% bins but reveal a zero-return spike and rounded tick clusters at 0.1% bins. Both views describe different resolutions.

PYTHON / SYSTEM IMPLEMENTATION

Anchor bin edges deterministically, label count versus density, include underflow and overflow, and reuse edges for honest group comparison. Render a rug or ECDF when n is small.

CONFUSIONS TO AVOID

Do not infer a probability density from counts with unequal bin widths. Auto-ranging each group separately makes tails look comparable when their numeric scales differ.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploration tools expose bin controls while saved reports record exact edges. Automated checks flag mass at zero, round-number heaping, and tail bins with suspicious vendor concentrations.

RELATED CONCEPTS AND QUICK RECALL

Related: density, bin width, discretization, ECDF. Recall task: Explain how using different bin edges can create or hide a return-distribution mode.

Kernel density and smoothing

BEGINNER

ONE-SENTENCE MEANING

Kernel density estimation creates a smooth distribution estimate by placing a small kernel around every observation. Bandwidth controls the tradeoff between revealing local structure and suppressing sampling noise.

MEMORY JOGGER

Histogram bins are replaced by overlapping bumps. Bandwidth, not artistic preference, determines how much detail survives.

WHY IT MATTERS IN QUANT RESEARCH

Density plots help compare continuous feature or return shapes, especially when overlaid carefully. They can expose regime mixtures, but should not replace tail counts or exact threshold probabilities.

FORMULA INTUITION

$\hat{f}_h(x) = (1/(nh)) \sum K((x-x_i)/h)$. Smaller h yields a rougher estimate; larger h can merge distinct modes and extend mass beyond feasible boundaries.

SIMPLE MARKET EXAMPLE

A bimodal intraday-return sample may reflect quiet and news regimes. Excess smoothing shows one broad mound, while too little bandwidth creates many meaningless peaks.

PYTHON / SYSTEM IMPLEMENTATION

Record kernel, bandwidth rule, weights, boundary correction, and sample size. Validate modes across nearby bandwidths and show the underlying rug or quantiles.

CONFUSIONS TO AVOID

Smooth curves can imply impossible negative spreads or leverage, and equal visual area can hide different sample sizes. KDE tail shape is weakly supported beyond dense observations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research visualization code treats KDE as an exploratory layer and stores no synthetic density as raw data. Regime classification requires validation beyond a visually appealing number of peaks.

RELATED CONCEPTS AND QUICK RECALL

Related: bandwidth, multimodality, mixture distribution, histogram. Recall task: How would you decide whether two KDE peaks are stable regimes or bandwidth artifacts?

Autocorrelation as a descriptive pattern

BEGINNER

ONE-SENTENCE MEANING

Autocorrelation measures linear association between a series and a lagged version of itself. It describes persistence or reversal at specific lags without proving a stable forecasting mechanism.

MEMORY JOGGER

Shift the series, align valid pairs, and compare today with the past. Positive values indicate persistence; negative values indicate alternation at that lag.

WHY IT MATTERS IN QUANT RESEARCH

Return, volatility, spread, order-flow, and signal autocorrelation affect effective sample size, execution assumptions, and feature horizon. The lag profile often matters more than lag one alone.

FORMULA INTUITION

$\rho_k = \text{Cov}(x_t, x_{t-k}) / \text{Var}(x_t)$ under a stable reference process. Sample autocorrelation uses only overlapping valid pairs and must state mean-removal and missing-value rules.

SIMPLE MARKET EXAMPLE

Raw daily returns may have near-zero lag-one correlation while absolute returns remain strongly autocorrelated, revealing volatility clustering rather than directional predictability.

PYTHON / SYSTEM IMPLEMENTATION

Compute an autocorrelation function with confidence bands chosen for the dependence context. Track pair count by lag and compare raw, absolute, squared, and residual series.

CONFUSIONS TO AVOID

A significant sample autocorrelation can arise from trend, seasonality, stale pricing, or multiple testing. Near-zero linear correlation does not establish independence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The diagnostics layer publishes lag tables for outcomes and model residuals. Validation uses the detected dependence to choose block lengths, standard errors, and evaluation splits.

RELATED CONCEPTS AND QUICK RECALL

Related: serial dependence, partial autocorrelation, effective sample size, volatility clustering. Recall task: Why can returns show little autocorrelation while squared returns remain strongly persistent?

Rolling windows and local statistics

BEGINNER

ONE-SENTENCE MEANING

A rolling statistic recomputes a summary over a moving fixed-length history, producing a local view that adapts as old observations exit and new ones enter. Window choice controls responsiveness and noise.

MEMORY JOGGER

A rolling window has a memory with a hard edge. Every update adds one observation and forgets one observation completely.

WHY IT MATTERS IN QUANT RESEARCH

Rolling means, volatility, quantiles, and correlations reveal changing conditions and support point-in-time features. They also create overlapping outputs that are highly dependent.

FORMULA INTUITION

For width w , $m_t = (1/w) \sum_{i=0}^{w-1} x_{t-i}$. The effective horizon depends on observation frequency, missing sessions, and whether the current value is included.

SIMPLE MARKET EXAMPLE

A 20-day volatility estimate reacts faster to a shock than a 252-day estimate but is noisier. A one-day lag determines whether it is available before today's trading decision.

PYTHON / SYSTEM IMPLEMENTATION

Declare window in observations or elapsed time, inclusion side, minimum periods, lag, calendar, and null policy. Unit-test values at boundaries and around missing dates.

CONFUSIONS TO AVOID

Centered windows leak future data, and silently shrinking windows make early history incomparable. Overlapping rolling metrics should not be treated as independent evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature registry versions every temporal parameter and uses event-time execution. Backtest code verifies feature timestamps precede order timestamps under the same calendar.

RELATED CONCEPTS AND QUICK RECALL

Related: look-ahead bias, expanding window, EWMA, overlapping data. Recall task: Design a point-in-time 60-day rolling mean and state exactly which date's return is the newest allowed input.

Expanding windows and cumulative evidence

BEGINNER

ONE-SENTENCE MEANING

An expanding statistic uses all eligible history from a fixed start through the current time. It gains stability as observations accumulate but adapts slowly when the process changes.

MEMORY JOGGER

Rolling forgets; expanding remembers. The extra stability is useful only if old evidence remains relevant.

WHY IT MATTERS IN QUANT RESEARCH

Expanding means, variances, and hit rates provide audit-friendly cumulative histories and reveal how conclusions developed. Comparison with rolling estimates is a direct drift diagnostic.

FORMULA INTUITION

At time t , $m_t = (1/n_t) \sum_{i \leq t} x_i$ over eligible history. Online variance can be updated with numerically stable algorithms without rereading the entire dataset.

SIMPLE MARKET EXAMPLE

An expanding Sharpe estimate may remain positive long after a strategy stops working because profitable early years dominate. A rolling estimate can expose the recent deterioration.

PYTHON / SYSTEM IMPLEMENTATION

Freeze start-date rationale, inclusion policy, and structural-break resets. Persist sufficient statistics and reconcile online updates against batch recomputation.

CONFUSIONS TO AVOID

More history does not guarantee a better current estimate under nonstationarity. Resetting the start after seeing poor performance is selection and must be governed.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Monitoring shows expanding, rolling, and regime-conditioned views together. Model governance permits resets only through versioned change events with preserved pre-reset history.

RELATED CONCEPTS AND QUICK RECALL

Related: online statistics, concept drift, rolling window, structural break. Recall task: What pattern between expanding and rolling mean would suggest a strategy's historical edge is stale?

Exponentially weighted statistics

BEGINNER

ONE-SENTENCE MEANING

Exponentially weighted statistics assign geometrically decreasing influence to older observations. They adapt smoothly without the hard cutoff of a rolling window, while retaining a clearly parameterized memory.

MEMORY JOGGER

Recent observations matter most, and influence fades rather than disappearing at one boundary. The decay rate is the model's memory.

WHY IT MATTERS IN QUANT RESEARCH

EW means and variances are common for volatility, covariance, signal smoothing, and live monitoring. They are responsive but can chase transient shocks when decay is too fast.

FORMULA INTUITION

$m_t = (1 - \lambda)x_t + \lambda m_{t-1}$. Half-life h implies $\lambda = 2^{(-1/h)}$; weight after h periods is half the current weight before normalization details.

SIMPLE MARKET EXAMPLE

With a 20-day half-life, a return from 20 sessions ago receives half the relative weight of today's return. A volatility shock therefore fades gradually.

PYTHON / SYSTEM IMPLEMENTATION

Store λ or half-life, initialization, adjustment convention, minimum history, and lag. Compare recursive output with a direct weighted calculation on fixed examples.

CONFUSIONS TO AVOID

Different software defaults for adjusted versus recursive EW calculations can disagree early in a series. A half-life is not the same as an equal-weight window length.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The online feature engine uses stable recursive states with checkpointing. Drift dashboards disclose effective memory and prevent parameter changes from rewriting historical features silently.

RELATED CONCEPTS AND QUICK RECALL

Related: decay, half-life, EWMA volatility, online update. Recall task: Translate a 10-day half-life into λ and explain how initialization affects the first estimates.

Realized volatility and range estimators

BEGINNER

ONE-SENTENCE MEANING

Realized volatility summarizes return variation over a period using intraperiod observations, while range-based estimators extract information from highs and lows. Both require clean prices and a precise sampling convention.

MEMORY JOGGER

Volatility is accumulated movement, not just endpoint change. Higher-frequency paths reveal variation hidden by a close-to-close return.

WHY IT MATTERS IN QUANT RESEARCH

Realized measures support risk forecasts, regime labels, execution sizing, and volatility-targeted portfolios. Microstructure noise and overnight gaps determine which estimator is appropriate.

FORMULA INTUITION

Realized variance $RV = \sum r_{t,j}^2$ within a day; realized volatility $= \sqrt{RV}$. Parkinson variance is proportional to $[\ln(H/L)]^2$ under restrictive continuous-price assumptions.

SIMPLE MARKET EXAMPLE

A day can open and close unchanged while moving widely intraday, producing zero close return but high realized volatility. A stop or option position experiences the path, not just the endpoint.

PYTHON / SYSTEM IMPLEMENTATION

Specify sampling interval, session, overnight treatment, timezone, bad-tick filters, and price source. Compare multiple frequencies to detect noise-induced variance inflation.

CONFUSIONS TO AVOID

Sampling faster is not automatically better because bid-ask bounce inflates squared returns. Range formulas can fail around jumps, price limits, and asynchronous high-low construction.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The volatility service publishes close, intraday, overnight, and robust realized components separately. Risk engines choose the component aligned with actual holding and trading hours.

RELATED CONCEPTS AND QUICK RECALL

Related: quadratic variation, high-low range, microstructure noise, volatility. Recall task: Explain how a flat close-to-close day can still generate high realized risk and trading costs.

Drawdown depth, duration, and recovery

BEGINNER

ONE-SENTENCE MEANING

A drawdown is the percentage decline of a wealth index from its prior running peak. Depth measures loss magnitude, duration measures time underwater, and recovery marks the return to a new peak.

MEMORY JOGGER

Drawdown asks how far and how long capital stayed below its best prior level. It is path dependent and cannot be reconstructed from an unordered return sample.

WHY IT MATTERS IN QUANT RESEARCH

Investors experience funding pressure, risk-limit breaches, and behavioral stress through drawdowns. Two strategies with equal mean and volatility can have very different underwater paths.

FORMULA INTUITION

Wealth $W_t = \text{product}_{\{i \leq t\}} (1 + r_i)$; running peak $M_t = \max_{\{s \leq t\}} W_s$; drawdown $D_t = W_t / M_t - 1$. Maximum drawdown is $\min D_t$.

SIMPLE MARKET EXAMPLE

A strategy falling 12% and recovering in ten days differs operationally from one down 8% for a year. Depth alone misses capital lockup and model-governance pressure.

PYTHON / SYSTEM IMPLEMENTATION

Compute from reconciled net returns with explicit cash flows and session order. Store peak date, trough date, recovery date, depth, duration, and unresolved status for every episode.

CONFUSIONS TO AVOID

Maximum drawdown is one realized path statistic, not a stable population parameter. Combining strategies can alter drawdowns through timing, and missing return days can hide the true trough.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The portfolio monitor tracks active and historical episodes against limits and recovery expectations. Scenario engines resample paths rather than merely shuffling summary statistics.

RELATED CONCEPTS AND QUICK RECALL

Related: wealth index, underwater curve, path dependence, recovery time. Recall task: Why does maximum drawdown require ordered returns, and what additional information does duration provide?

Downside, upside, and semideviation

BEGINNER

ONE-SENTENCE MEANING

Downside and upside summaries split outcomes relative to a threshold, while semideviation measures dispersion on one side. They describe directional risk that symmetric standard deviation deliberately merges.

MEMORY JOGGER

Choose a target, then ask how often and how far outcomes miss or exceed it. Threshold choice is part of the metric, not a hidden constant.

WHY IT MATTERS IN QUANT RESEARCH

Downside deviation, gain-loss ratios, and capture statistics help distinguish defensive strategies from leveraged carry. They align measurement with asymmetric investor losses and objectives.

FORMULA INTUITION

Downside semivariance about target τ is $\text{mean}[\min(x_i - \tau, 0)^2]$, often square-rooted. Conditional downside mean averages only observations below τ and answers a different question.

SIMPLE MARKET EXAMPLE

Two strategies may each have 1% standard deviation, but one concentrates variation above zero while the other varies mostly through losses. Downside deviation separates those profiles.

PYTHON / SYSTEM IMPLEMENTATION

Declare threshold units and horizon, denominator convention, and whether non-downside observations contribute zeros or are excluded. Report the below-threshold count.

CONFUSIONS TO AVOID

Using only negative observations changes the denominator and can make cross-sample comparisons deceptive. A zero threshold and a required-return threshold produce different economic interpretations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Risk scorecards pair symmetric volatility with downside deviation, expected shortfall, and threshold frequency. Objective configuration is centralized so teams cannot tune τ to improve presentation.

RELATED CONCEPTS AND QUICK RECALL

Related: Sortino ratio, lower partial moment, target return, asymmetry. Recall task: Contrast downside deviation around zero with downside deviation around a 5% annual target.

Seasonality and intraday profiles

BEGINNER

ONE-SENTENCE MEANING

Seasonality is systematic variation associated with calendar or clock positions such as weekday, month, or minutes since the open. A profile summarizes repeated observations at the same position in a cycle.

MEMORY JOGGER

Group by a recurring clock, but preserve the date as the independent event. A pattern must repeat across cycles rather than rely on many rows from one day.

WHY IT MATTERS IN QUANT RESEARCH

Intraday volume, spread, volatility, and return profiles guide execution and feature normalization. Calendar patterns can also expose data timestamp errors or market-structure changes.

FORMULA INTUITION

A slot mean is $\sum_d x_{\{d,s\}} / n_s$ for dates d at slot s . Robust profiles use medians or quantiles, and uncertainty should cluster by date when within-day slots share shocks.

SIMPLE MARKET EXAMPLE

Equity spreads often widen near the open, narrow midday, and change into the close. Comparing a 10:00 spread with the all-day mean confounds time-of-day structure with abnormality.

PYTHON / SYSTEM IMPLEMENTATION

Normalize timestamps to exchange session, handle daylight saving and half-days, and report date count per slot. Compare profile stability by year, venue, and volatility regime.

CONFUSIONS TO AVOID

Pooling pre- and post-market data with regular hours creates artificial modes. Thousands of minute rows are not thousands of independent days of seasonal evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar service assigns canonical session slots, and execution analytics use versioned baseline profiles. Monitoring alerts on deviations after accounting for expected clock position.

RELATED CONCEPTS AND QUICK RECALL

Related: calendar effects, intraday seasonality, clustering, execution benchmark. Recall task: Design a diagnostic that distinguishes a genuine opening-spread pattern from one abnormal day with many quotes.

Regime-conditioned summaries

BEGINNER

ONE-SENTENCE MEANING

A regime-conditioned summary computes statistics within explicitly defined market states rather than averaging across all history. It reveals heterogeneity but depends on how states are labeled and when labels become known.

MEMORY JOGGER

Overall averages blend environments. Split only on state definitions that are observable, reproducible, and supported by enough data.

WHY IT MATTERS IN QUANT RESEARCH

Signal return, turnover, correlation, and tail behavior often differ in quiet, trending, stressed, or illiquid periods. Conditional summaries inform sizing, fallback logic, and robustness expectations.

FORMULA INTUITION

For state z , mean $\mu_z = \sum I(Z_i=z)x_i / \sum I(Z_i=z)$. The unconditional mean equals the state means weighted by state frequencies, which can change through time.

SIMPLE MARKET EXAMPLE

A strategy may earn 8 bp in calm periods and lose 30 bp in crises yet show a positive overall mean because crises are rare. Capital planning must respect the conditional loss.

PYTHON / SYSTEM IMPLEMENTATION

Define labels point-in-time, show counts and transition dates, and test neighboring thresholds. Separate ex-ante deployable states from ex-post explanatory labels.

CONFUSIONS TO AVOID

Using future realized volatility to label today's regime leaks outcomes. Fine partitions create unstable summaries and invite cherry-picking of favorable states.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The regime service versions definitions and availability timestamps. Research artifacts display conditional metrics plus the frequency weights that reconstruct the aggregate.

RELATED CONCEPTS AND QUICK RECALL

Related: mixture distribution, structural break, conditional expectation, state model. Recall task: How can a positive unconditional mean coexist with a loss in the state that matters most for risk?

Cross-sectional ranks and percent ranks

BEGINNER

ONE-SENTENCE MEANING

A cross-sectional rank orders assets within the same timestamp and eligible universe. Percent rank maps that order to a relative scale, reducing sensitivity to raw units and monotone transformations.

MEMORY JOGGER

Rank answers who is high or low now, not how unusual an asset is relative to its own history.

WHY IT MATTERS IN QUANT RESEARCH

Factor portfolios, relative-value signals, and breadth indicators frequently use ranks because assets have different scales. Rank stability depends on universe membership and tie handling.

FORMULA INTUITION

For N assets, a common percent rank is $(\text{rank}-1)/(N-1)$, with conventions for ties and endpoints. Quantile buckets partition ordered assets into groups of approximately equal count.

SIMPLE MARKET EXAMPLE

A 3% return can rank first in a quiet market and fiftieth in a broad rally. The same raw value therefore produces a different relative signal.

PYTHON / SYSTEM IMPLEMENTATION

Construct the point-in-time universe first, define ascending direction, tie method, null policy, and minimum N . Record ranks before downstream neutralization or portfolio constraints.

CONFUSIONS TO AVOID

Ranks erase magnitude: the gap between first and second may be tiny or enormous. Changing universe coverage can move every rank without any asset value changing.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The cross-sectional feature engine binds ranks to universe snapshots and publishes membership counts. Portfolio construction receives both rank and raw value to diagnose crowding and discontinuity.

RELATED CONCEPTS AND QUICK RECALL

Related: quantile portfolio, ordinal data, universe drift, normalization. Recall task: Contrast a time-series percentile with a same-day cross-sectional percentile for momentum.

Breadth and cross-sectional dispersion

BEGINNER

ONE-SENTENCE MEANING

Breadth measures how widely a market move or signal is shared, while cross-sectional dispersion measures how spread out asset outcomes are at one time. Together they distinguish broad participation from concentrated leadership.

MEMORY JOGGER

Index direction says where the aggregate moved; breadth says how many joined; dispersion says how different members were.

WHY IT MATTERS IN QUANT RESEARCH

Breadth and dispersion inform regime context, stock-selection opportunity, hedging quality, and index concentration. A flat index can conceal large offsetting moves with rich relative-value opportunity.

FORMULA INTUITION

Breadth may be $\text{fraction}(r_i > 0)$ or fraction above a benchmark. Dispersion can be cross-sectional standard deviation, IQR, or median absolute deviation at date t .

SIMPLE MARKET EXAMPLE

An index gains 1% while only 30% of members rise because a few mega-caps dominate. Positive index return with weak breadth signals concentrated participation.

PYTHON / SYSTEM IMPLEMENTATION

Use point-in-time membership and report equal- and value-weighted breadth. Compute robust dispersion, constituent count, and concentration measures on the same snapshot.

CONFUSIONS TO AVOID

Breadth depends on the threshold and universe; stale prices can create false zeros. Value-weighted breadth can become another index-return proxy rather than a participation measure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The market-state layer publishes breadth, dispersion, concentration, and coverage together. Strategy monitors use them as context, not standalone entry signals without validation.

RELATED CONCEPTS AND QUICK RECALL

Related: advance-decline, concentration, factor opportunity, cross-section. Recall task: Describe a market day with positive index return, weak breadth, and high dispersion, and name one portfolio implication.

Correlation and covariance summaries

BEGINNER

ONE-SENTENCE MEANING

Covariance measures joint variation in original units, while correlation standardizes covariance to a -1 to 1 linear-association scale. A matrix summarizes pairwise dependence but can hide nonlinear and state-specific relationships.

MEMORY JOGGER

Covariance carries scale and feeds portfolio variance; correlation removes scale and supports comparison. Neither establishes causation or stability.

WHY IT MATTERS IN QUANT RESEARCH

Descriptive dependence maps identify clusters, redundant features, diversification assumptions, and common shocks. Rolling and regime-conditioned versions show whether the average matrix is representative.

FORMULA INTUITION

$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$; $\rho = \text{Cov} / (\sigma_X \sigma_Y)$. Portfolio variance is $w' \Sigma w$, making covariance units economically operative.

SIMPLE MARKET EXAMPLE

Two assets with 0.8 correlation can still contribute differently to risk if one is four times as volatile. Covariance and weights determine the actual portfolio interaction.

PYTHON / SYSTEM IMPLEMENTATION

Align timestamps and horizons, require pair counts, handle asynchronous markets, and apply positive-semidefinite checks before optimization. Show clustered heatmaps with a fixed color scale.

CONFUSIONS TO AVOID

Pairwise deletion can create an incoherent matrix, and correlation near zero can coexist with strong tail dependence. Short rolling windows produce noisy extreme values.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk data product versions the estimation window, missingness policy, shrinkage, and universe. Research dashboards expose pair counts and state sensitivity beside the matrix.

RELATED CONCEPTS AND QUICK RECALL

Related: portfolio variance, tail dependence, clustering, shrinkage. Recall task: Why is a correlation heatmap insufficient for deciding how much diversification two assets provide?

Missingness and coverage summaries

BEGINNER

ONE-SENTENCE MEANING

Missingness summaries quantify where expected observations are absent, while coverage measures how much of the target frame is observed and usable. Their patterns often carry more information than one overall null percentage.

MEMORY JOGGER

Count what should have existed, not only null cells in rows that arrived. Missing rows and missing values are different failure modes.

WHY IT MATTERS IN QUANT RESEARCH

Market halts, stale quotes, vendor outages, delistings, and illiquidity can make data absence economically selective. Coverage bias can distort returns, spreads, and model evaluation.

FORMULA INTUITION

Coverage=observed eligible units/expected eligible units. Summarize by date, asset, venue, field, state, and consecutive-gap length to reveal structure.

SIMPLE MARKET EXAMPLE

A feed with 99% overall coverage may lose 40% of quotes during the ten minutes around a volatile open. The missing 1% is precisely the period most relevant to execution risk.

PYTHON / SYSTEM IMPLEMENTATION

Build expected calendars and universe grids, classify absence reasons, and distinguish null, stale, zero, and not-applicable states. Alert on clustered gaps and denominator changes.

CONFUSIONS TO AVOID

Forward filling can convert missingness into misleading stability, while dropping incomplete assets creates survivor-like selection. Zero volume is not the same as unknown volume.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Data observability dashboards run before research metrics and publish quality states. Feature jobs fail closed or use documented fallbacks based on field criticality.

RELATED CONCEPTS AND QUICK RECALL

Related: data quality, stale values, missing mechanism, sampling frame. Recall task: Design a coverage table that would reveal a short exchange-open outage hidden by a 99% daily average.

Outliers and influence diagnostics

BEGINNER

ONE-SENTENCE MEANING

An outlier is an observation far from a reference pattern, while influence measures how much a statistic or model changes because of that observation. Extreme does not mean erroneous, and moderate points can still be influential.

MEMORY JOGGER

First ask whether the value is valid; then ask how much the conclusion depends on it. Detection, diagnosis, and treatment are separate steps.

WHY IT MATTERS IN QUANT RESEARCH

Bad ticks can manufacture alpha, but genuine crashes define tail risk. Leave-one-event and contribution analyses show whether a reported edge rests on a single asset or date.

FORMULA INTUITION

Mean contribution is x_i/n , variance contribution is $(x_i - \bar{x})^2$, and leave-one-out influence compares $S(\text{data})$ with $S(\text{data without } i)$. Robust scores use median and MAD.

SIMPLE MARKET EXAMPLE

One acquisition jump can create most of a factor's annual return. If the event is valid, the research question becomes whether that concentrated payoff is repeatable and tradable.

PYTHON / SYSTEM IMPLEMENTATION

Join flags to source records, corporate actions, news, venue, and quality lineage. Report top contributors to mean, variance, skew, and drawdown with before-and-after sensitivity.

CONFUSIONS TO AVOID

Automatic deletion biases tails and can improve backtests selectively. A standard z-score can underflag an extreme because that same observation inflates mean and standard deviation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quality system owns correction of invalid records; research owns robustness to valid extremes. Every adjustment preserves originals and a reason code.

RELATED CONCEPTS AND QUICK RECALL

Related: leverage, robust statistics, data error, sensitivity analysis. Recall task: Give one example of a valid outlier and one invalid outlier, and describe different system responses.

Grouped summaries and Simpson's paradox

BEGINNER

ONE-SENTENCE MEANING

Grouped summaries describe outcomes within categories, while Simpson's paradox occurs when an aggregate association reverses or changes after accounting for group composition. Weight shifts can move a headline metric even when every group is unchanged.

MEMORY JOGGER

The total is a weighted blend of groups. Before explaining movement, separate within-group change from changing group weights.

WHY IT MATTERS IN QUANT RESEARCH

Venue mix, asset liquidity, sector composition, and volatility regimes can alter average slippage or returns. Aggregate improvement may reflect trading easier names rather than better execution.

FORMULA INTUITION

Aggregate mean = $\sum p_g \mu_g$. Its change decomposes into within-group effects, mix effects, and an interaction term depending on the chosen reference weights.

SIMPLE MARKET EXAMPLE

Desk A can have lower slippage than Desk B in both liquid and illiquid stocks yet worse overall slippage because A handles far more illiquid flow.

PYTHON / SYSTEM IMPLEMENTATION

Publish group counts, economic weights, within-group metrics, and standardized totals under fixed reference weights. Use consistent group boundaries and track migration between buckets.

CONFUSIONS TO AVOID

Overgrouping creates noisy cells, while using current information to define historical groups leaks future state. Aggregate comparisons without composition tables invite false causal stories.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Performance dashboards include a mix-adjusted bridge before operational attribution. Metric definitions identify the approved grouping hierarchy and reference-weight version.

RELATED CONCEPTS AND QUICK RECALL

Related: composition effect, standardization, attribution, aggregation bias. Recall task: Construct a two-group example where every subgroup improves but the overall average worsens because weights shift.

Descriptive metric contracts

BEGINNER

ONE-SENTENCE MEANING

A metric contract is a versioned specification of a summary's business meaning, formula, input grain, eligibility, weights, units, window, and quality rules. It turns a notebook number into a reproducible system object.

MEMORY JOGGER

If two engineers cannot independently produce the same numerator and denominator, the metric is not defined well enough for governance.

WHY IT MATTERS IN QUANT RESEARCH

Quant systems reuse statistics across research, risk, and monitoring. Contracts prevent semantic drift such as gross versus net return, calendar versus trading days, or equal versus capital weighting.

FORMULA INTUITION

A complete contract defines $\text{value} = (\text{numerator}) / (\text{denominator})$ or an exact algorithm, with temporal cutoff, null behavior, grouping, annualization, precision, and uncertainty metadata.

SIMPLE MARKET EXAMPLE

A dashboard labeled average slippage can differ by fill weighting, arrival-price timestamp, fees, currency conversion, and canceled-order treatment. Each choice changes the number.

PYTHON / SYSTEM IMPLEMENTATION

Assign stable IDs, owners, tests, schema dependencies, and semantic versions. Include hand-calculated fixtures, reconciliation tolerances, and backward-compatibility policy.

CONFUSIONS TO AVOID

A SQL query alone is not a definition because table semantics and eligibility may change. Display rounding can conceal material differences near limits.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric registry generates consistent research and monitoring implementations, while lineage links each rendered value to code, data snapshot, and contract version.

RELATED CONCEPTS AND QUICK RECALL

Related: data contract, lineage, semantic versioning, reconciliation. Recall task: Write the minimum contract fields needed for a 20-day annualized volatility metric.

Descriptive dashboards and monitoring

BEGINNER

ONE-SENTENCE MEANING

A descriptive dashboard organizes level, distribution, trend, segmentation, and data-quality views around a decision. It should reveal denominators and uncertainty rather than presenting polished point estimates without context.

MEMORY JOGGER

Start with what changed, then show where, how broad, how unusual, and whether the data are trustworthy. Every chart should support an action or diagnosis.

WHY IT MATTERS IN QUANT RESEARCH

Research and live systems need a shared language for returns, risk, costs, features, and coverage. Monitoring detects drift only when historical and live calculations use identical contracts.

FORMULA INTUITION

A useful scorecard pairs current value, reference period, delta, sample count, coverage, robust range, and control limits. Distribution and segment panels explain the headline rather than repeat it.

SIMPLE MARKET EXAMPLE

A jump in average slippage may come from wider within-asset costs, a shift toward illiquid names, one extreme order, or missing easy fills. The dashboard should separate these paths.

PYTHON / SYSTEM IMPLEMENTATION

Freeze time filters, show freshness and quality state, enable drilldown to observations, and version all thresholds. Test empty, partial, extreme, and late-data states before release.

CONFUSIONS TO AVOID

Red-green coloring without economic thresholds encourages overreaction, and auto-scaled axes can exaggerate noise. A dense dashboard is not useful if panels answer no defined question.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The observability layer consumes registry metrics, posts alerts with lineage, and routes owners to diagnostic views. Research reports reuse the same definitions to close the backtest-to-live gap.

RELATED CONCEPTS AND QUICK RECALL

Related: control chart, observability, data lineage, metric registry. Recall task: Design the diagnostic sequence for an alert that average live slippage rose from 4 to 7 basis points.

From raw prices to valid summaries

BEGINNER

DESIGN RULE

Every summary inherits the correctness of the price lineage, entity alignment, calendar, corporate-action policy, return horizon, and eligibility frame that precede it.

1

Raw market events

Trades, quotes, bars, reference data, vendor timestamps

2

Canonical observations

Entity mapping, timezone, session, currency, duplicate policy

3

Economic price series

Raw execution prices plus adjusted and total-return variants

4

Return construction

Simple, log, excess, horizon, lag, gap and stale-value rules

5

Eligibility snapshot

Point-in-time universe, coverage, quality state, exclusions

6

Descriptive layer

Center, spread, quantiles, shape, time, cross-section, groups

7

Decision artifact

Feature, risk limit, research report, dashboard, alert with lineage

AUDIT CHECKPOINT

Reconcile one asset and one session by hand from raw price through final statistic. Store numerator, denominator, and excluded-record ledger beside the result.

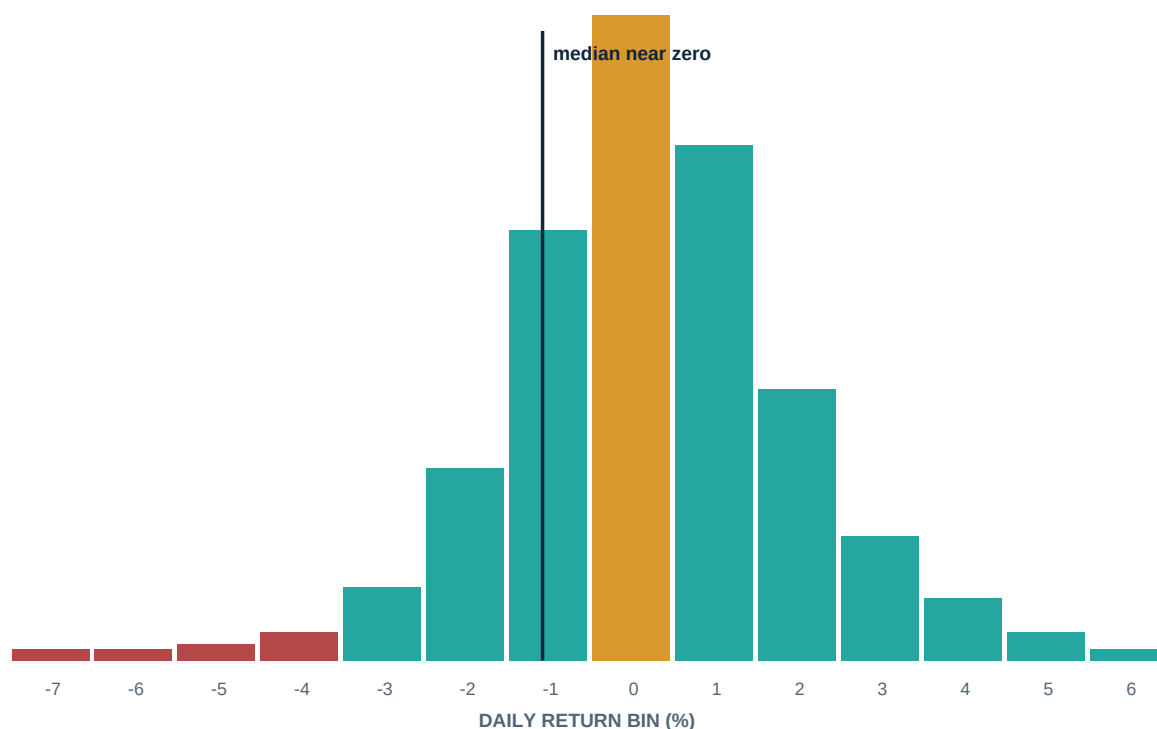
Reading a market-return distribution

BEGINNER

SYNTHETIC SAMPLE

The chart combines a quiet core, a zero-return spike, modest positive skew near the center, and a few severe left-tail events. Each descriptive element answers a different question.

COUNT



INTERPRETATION STACK

Center: typical direction and economic expectation. Spread: ordinary variability. Shape: asymmetry and clustering. Tails: rare but capital-relevant loss magnitude. Mass at zero: possible staleness, price grid, or genuine inactivity.

DO NOT COMPRESS TOO SOON

A mean and standard deviation would conceal the zero spike and left-tail concentration. Add quantiles, exceedance counts, skew, robust scale, and event contribution before choosing a distribution model.

VALIDATION

Redraw with nearby bin widths, inspect the ECDF, and compare by volatility regime. Trace the three worst observations to raw records and known events before classifying them as risk or error.

Rolling, expanding, and exponential memory

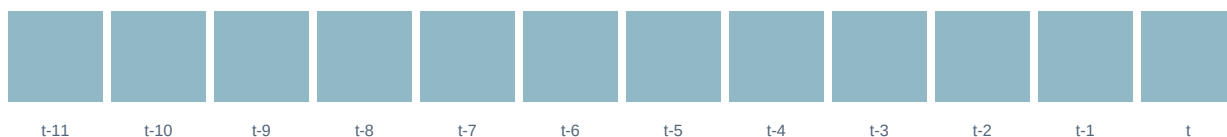
BEGINNER

SAME HISTORY, DIFFERENT MEMORY

The estimator is not defined by its statistic alone. The reference window determines which past observations influence today's value and by how much.

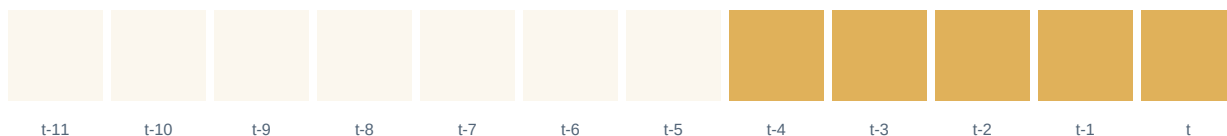
Expanding

All available history; stable but slow to forget stale regimes



Rolling 5

Equal weight inside a fixed window; hard cutoff at t-5



EW half-life 3

Smooth decay; every observation remains but recent ones dominate



POINT-IN-TIME REQUIREMENT

For a decision made before today's close, the newest valid input may be t-1. Shifting after the rolling calculation is not equivalent to shifting inputs first when missingness or grouping changes.

METHOD SELECTION

Use expanding summaries for cumulative evidence, rolling windows for bounded local context, and exponential weights for smooth adaptation. Compare conclusions across plausible memory choices instead of optimizing one window on backtest performance.

Conditional performance and mixture risk

BEGINNER

READING RULE

The same strategy can have different location, dispersion, and tails across market states. Counts and state frequencies are as important as the colored metric values.

	Calm	Trend	Choppy	Stress
Mean bp	7	14	-3	-28
Vol bp	38	65	91	210
5th pct bp	-52	-81	-142	-390
Win rate %	57	61	48	33
Days	410	225	175	38

INTERPRETATION

Calm and trending states support the headline return, while the thin stress sample contains a much larger average loss and tail. The unconditional mean is a frequency-weighted mixture, not a universal strategy property.

VALIDATION

Check whether state labels were observable before trading, whether thresholds are stable, and whether stress observations represent distinct events. Report uncertainty rather than treating 38 days as a precise regime law.

SYSTEM ACTION

Use the table to set conditional exposure caps, monitoring expectations, and fallback triggers. Do not activate a regime switch until the label pipeline, delay, and transition behavior are tested point-in-time.

Choosing a descriptive view

BEGINNER

START FROM THE QUESTION

No single summary is best. Pair a primary statistic with a diagnostic that exposes its most likely failure mode.

QUESTION	PRIMARY VIEW	PAIR WITH	WHY
Expected P&L	Mean; weighted mean	Tail contribution	Capital adds outcomes
Typical observation	Median; quantiles	Mean-median gap	Robust to extremes
Ordinary variability	Std dev plus MAD	SD/MAD ratio	Detect tail dominance
Downside risk	Lower quantiles; ES	Exceedance count	Direction matters
Distribution comparison	ECDF	Tail threshold gaps	Avoid bin dependence
Local market state	Rolling or EW stats	Memory sensitivity	Point-in-time adaptation
Relative asset position	Cross-sectional rank	Raw magnitude	Scale-free ordering
Execution quality	Weighted cost plus quantiles	Mix-adjusted groups	Volume and tails
Data reliability	Coverage and staleness	Gap clusters	Denominator first

ESCALATION RULE

When robust and ordinary summaries disagree materially, or aggregate and grouped views tell different stories, investigate the observations and composition before publishing a causal explanation.

Formula sheet and interpretation map

BEGINNER

SIMPLE AND LOG RETURN

$r_t = P_t/P_{t-1} - 1$; $\ell_t = \ln(1+r_t)$. Simple returns compound multiplicatively; log returns add through time.

MEAN AND WEIGHTED MEAN

$\bar{x} = \sum x_i/n$; $\bar{x}_w = \sum w_i x_i / \sum w_i$. The weights define the represented economic perspective.

SAMPLE VARIANCE

$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$. Squared deviations make extremes influential; $n-1$ does not cure tails or dependence.

QUANTILES AND IQR

Q_q is the q -position of the ordered sample; $IQR = Q_{0.75} - Q_{0.25}$. Pin the interpolation convention.

ROBUST SCALE

$MAD = \text{median}(|x_i - \text{median}(x)|)$. Multiply by 1.4826 only when a normal-reference sigma estimate is intended.

SKEW AND KURTOSIS

$\gamma_1 = E[(X - \mu)^3] / \sigma^3$; $\gamma_2 = E[(X - \mu)^4] / \sigma^4$. Cubic direction and fourth-power extremes drive interpretation.

EW MEAN

$m_t = (1 - \lambda)x_t + \lambda m_{t-1}$; $\lambda = 2^{-(1/h)}$ for half-life h . Initialization and lag are part of the definition.

DRAWDOWN

$W_t = \text{product}(1+r_i)$; $D_t = W_t / \max_{(s \leq t)} W_s - 1$. Ordering and net-return reconciliation are mandatory.

Worked calculation: return sample

BEGINNER

1. SAMPLE AND UNITS

Five daily net returns are -4.0%, -1.0%, 0.5%, 1.0%, and 2.5%. The rows are one strategy-day, equally weighted, ordered only for quantiles.

2. MEAN AND MEDIAN

The sum is -1.0%, so mean=-0.20%. The middle value is 0.50%, so median=0.50%. Their sign disagreement reveals a loss-dominated tail.

3. COMPOUNDED RETURN

Wealth=(0.96)(0.99)(1.005)(1.01)(1.025)=approximately 0.9887, for cumulative return near -1.13%. Arithmetic sum is not compound performance.

4. QUANTILES AND IQR

Using median-of-halves for illustration, $Q1=(-4\%-1\%)/2=-2.5\%$ and $Q3=(1\%+2.5\%)/2=1.75\%$; IQR=4.25 percentage points. Software interpolation may differ.

5. SAMPLE DISPERSION

Deviations from -0.20% are -3.8, -0.8, 0.7, 1.2, and 2.7 percentage points. Squared deviations sum to 24.50, so $s=\sqrt{24.50/4}=2.47\%$.

6. DOWNSIDE VIEW

Two of five days are negative, but the -4% day contributes most of the total loss and variance. Win rate or median alone would give a misleadingly favorable impression.

7. ROBUSTNESS CHECK

Remove the -4% day only as a sensitivity diagnostic: the remaining mean becomes 0.75%. The 0.95 percentage-point shift shows the conclusion depends on one valid event.

8. DECISION RECORD

Store the raw dates, adjustment status, cost version, quantile convention, count, mean, median, standard deviation, cumulative return, and leave-one-out change before drawing a strategy conclusion.

Failure modes and diagnostic actions

BEGINNER

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Mean jumps; median stable	Tail event or bad tick	Top mean contributors	Validate source; keep valid tail
Volatility doubles; MAD flat	Few large moves	SD/MAD and event dates	Segment tail; test regime
Rank shifts; raw value flat	Universe coverage changed	Membership and N	Repair frame; annotate break
Slippage rises overall	Trade mix changed	Group weights and fixed mix	Publish mix-adjusted bridge
Rolling feature looks predictive	Look-ahead or overlap	Window endpoints; lag	Rebuild point-in-time
Zero-return mass increases	Staleness or price grid	Quote age and ticks	Classify inactivity vs missing
Correlation matrix unstable	Thin pairs or regime shift	Pair counts; state heatmap	Shrink or narrow scope
Dashboard is precise, feed thin	Hidden denominator loss	Coverage by time/group	Fail closed or show partial

TRIAGE ORDER

Confirm grain, units, eligibility, and data validity first. Then inspect weighting and grouping, distribution shape, time dependence, and only afterward the economic interpretation.

Final active recall and system index

BEGINNER

MEASUREMENT CONTRACT

Name observation, variable, unit, grain, entity, clock, horizon, eligibility, weight, and source without relying on column names.

RETURN LITERACY

Distinguish raw price, adjusted price, simple return, log return, cumulative return, dollar P&L, and annualized approximations.

CENTER AND SCALE

Use mean for economic aggregation, median for typical rank position, standard deviation for squared dispersion, and robust scale as a tail diagnostic.

SHAPE AND TAILS

Interpret quantiles, ECDF, histogram, skew, kurtosis, exceedance counts, and contribution rather than labeling every return distribution normal.

TIME STRUCTURE

Specify rolling, expanding, or exponential memory; lag features; inspect autocorrelation, seasonality, realized volatility, drawdown, and regimes.

CROSS-SECTION

Bind ranks, breadth, dispersion, and matrices to point-in-time membership, common timestamps, pair counts, and explicit weighting.

ROBUSTNESS

Preserve raw observations, separate valid extremes from errors, compare ordinary and robust summaries, and use leave-one-event sensitivity.

AGGREGATION

Expose group counts and weights, standardize mix when needed, and check for Simpson's paradox before attributing aggregate changes.

PRODUCTION STANDARD

Publish metric ID, formula, grain, units, window, lag, quality state, lineage, tests, owner, threshold, and diagnostic path.